Course Information

CHM 26200, Organic Chemistry, Spring 2022, 1:30 pm – 2:20 pm, MWF, Wetherill 200

On-line asynchronous: class sessions posted on Brightspace;

Instructor Contact Information

Instructor: Professor Jianguo Mei; **Room:** WTHR 465; **Phone:** 765-494-7156
Email: jgmei@purdue.edu (place CHM 26200 in the subject line of all emails)

Assistant Instructor: Dr. Vaidehi Pandit; **Room**: WTHR 465; **Email**: vpandit@purdue.edu

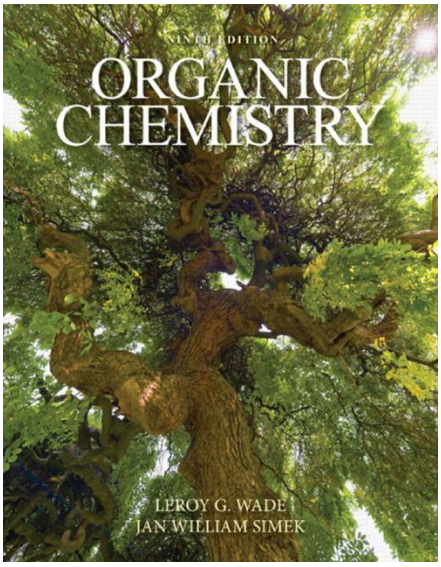
Course TAs: Atheena Jenkins; **Room:** WTHR 354; **Email:** jenkin94@purdue.edu
Joseph Romo; **Room:** WTHR 354; **Email**: romo8@purdue.edu

Office Hours: MWF, 3:30 – 4:30 pm via Zoom or WTHR 333C
https://zoom.us/j/8335838302?pwd=OS9RYkZmK2VpQ2hQWmFBTWZZTHFKdz09
or by appointment (Prof. Mei has an open-door policy)

Course Description

CHM 26200 is the second semester of a comprehensive, two-semester exploration of organic compounds, their chemistry, and mechanisms of transformation, stereochemical principles, structure-property analysis, target-oriented synthesis, and structure elucidation using different analytical techniques. In CHM 26100, you have learned the chemical principles underlying organic chemistry of aliphatic, halogenated, unsaturated and hydroxylated alkyl compounds. You have also been exposed to modern theory, stereochemistry, and characterization tools to illustrate the logic inherent in the subject matter and to demonstrate the predictability of many chemical transformations and their relevance to chemical engineering and materials science. In CHM 26200, you are expected to experience functional groups such ether, epoxide, ketone, aldehyde, carboxylic acid and their derivatives, as well as conjugated and aromatic functionality. You will learn to recognize recurring themes and patterns of reactivity of functional groups, develop an understanding of organic structure-property relationships, and apply an understanding of reaction mechanisms to new transformations of organic molecules. You will also explore catalytic carbon-carbon bond and carbon-heteroatom formation – the foundation of organic synthesis. At the end of the class, hopefully, you will recognize the role that organic chemistry plays as an enabling science in pharmaceutical, agricultural, electronic/photonic materials and commodity chemical industries

Learning Resources, Technology & Texts



- Brightspace page
  - You can access the course via Brightspace. It is strongly suggested that you explore and become familiar not only with the site navigation, but with content and resources available for this course. All course announcements and adaptations to potential disruptions due to COVID-19 will be found here.

- Textbook: **L.G. Wade and J.W. Simek, Organic Chemistry, 9th Edition**; available in hardcopy, loose-leaf (3-hole punched) or e-Text formats
  - Reading assignments and suggested practice problems will be based on this text.
  - We will cover approximately one chapter in Wade & Simek each week.
  - You may benefit from the use of molecular models that will allow you to build different organic molecules and examine.

- Other Textbook: **Brown, Iverson, Anslyn and Foote, Organic Chemistry, 7th Edition**. Chapter 24 – Catalytic Carbon-Carbon Bond Formation
  - Michigan State Virtual Textbook of Organic Chemistry: https://www2.chemistry.msu.edu/faculty/reusch/VirtTxtJml/intro1.htm
  - https://chem.libretexts.org/Bookshelves/Organic_Chemistry/Map%3A_Organic_Chemistry_(Wade)

- Other resources:
  - Many helpful organic chemistry lectures can be found at Khan Academy:
  - https://www.khanacademy.org/science/organic-chemistry

- Piazza: This semester, we will be using Piazza for class discussion. This tool is designed to get you help fast and efficiently from classmates, the TAs, and myself. Rather than emailing or texting questions to the teaching staff or other students, I encourage you to post them on Piazza to facilitate Q&A interactions.
  - https://piazza.com/purdue/spring2022/chm26200

- Model Kits: Organic molecules have complicated 3-dimensional structures that you need to be able to see and understand. It is often made easier if you have a 3-D model to examine and consider. Chem3D and ChemDraw are part of ChemOffice. Purdue has a site license for Chemoffice, which is a convenient molecular drawing program. You may find it handy for drawing molecular structures and view 3D configurations, and you are welcome to use it for the homework assignments.
    - https://guides.lib.purdue.edu/chemlibrary/software

Learning Outcomes

- Learn to recognize recurring themes and patterns of reactivity of functional groups
- Develop an understanding of organic structure-property relationships
- Use spectrometric and spectroscopic techniques to determine the structures of organic molecules
- Apply an understanding of reaction mechanisms to new transformations of organic molecules
- Recognize different classes of biomacromolecules, lipids, and synthetic polymers
- Recognize the role that organic chemistry plays as an enabling science in pharmaceutical, agricultural, electronic/photonic materials and commodity chemical industries

One of the most effective ways to learn organic chemistry and prepare for assessments is by solving the homework problems as soon as possible after reading the relevant text in Wade and its coverage in lecture to improve your comprehension and familiarity with the material. Additional exercises in the text are suggested below to help strengthen your understanding.

This course moves quickly and the >670 pages of detailed content build cumulatively over the course of the semester. If you are struggling in the course, you are strongly encouraged to act quickly and make an appointment to work with Prof. Mei, Ms. Jenkins, the Chemistry Resource Room, or a personal tutor before you fall too far behind. ***Developing the discipline to rigorously study the course materials at least two hours each day is your best assurance of success in this course.*** Most students who have difficulty with CHM 26200 do so because they lost focus in their studies for some period, not because the material was "too hard". Once they lost that momentum, they found it difficult to catch back up with the class due to its rigorous pace. Read and work problems every day!

Abbreviated Course Schedule

Schedule/assignment dates may change. Changes will be posted as Brightspace Announcements.

Week	Readings in Wade & Simek	Assignments
Week 1, Jan 10	Chap 15,	Homework 1 (Jan 14)
Week 2, Jan 17	Chap 16	
Week 3, Jan 24	Chap 17	Homework 2 (Jan 28)
Week 4, Jan 31	Chap 14	Assessment 1 (Feb 3) in **PHYS 112/203**
Week 5, Feb 7	Chap 18	Homework 3 (Feb 11)
Week 6, Feb 14	Chap 19	
Week 7, Feb 21	Chap 20	Homework 4 (Feb 25)
Week 8, Feb 28	Chap 21	Assessment 2 (Mar 3) in **LILY 1105**
Week 9, Mar 7	Chap 22	Homework 5 (Mar 11)
Week 10, Mar 14	Spring Break	
Week 11, Mar 21	Handout-1	Homework 6 (Mar 25)
Week 12, Mar 28	Handout-2	
Week 13, Apr 4	Handout-3	Assessment 3 (Apr 7) in **LILY 1105**
Week 14, Apr 11	Chap 26	
Week 15, Apr 18	Handout-4	Synthesis Problem (Apr 22)
Week 16, Apr 25	Handout-5	

Important Dates on the Academic Calendar

- Jan 10 – Classes Begin
- Jan 17 – MARTIN LUTHER KING JR. DAY (No Classes)
- Mar 14-19 – Spring Break
- Apr 30 – Classes End
- May 2-7 – Final Exams
- May 10 – Grades Due

For the full calendar, see: https://www.purdue.edu/registrar/calendars/2021-22-Academic-Calendar.html

Assignments

Your learning will be assessed through a combination of six biweekly homework sets, three open response assessments, an individualized synthesis project, and a comprehensive final assessment that are assigned according to a schedule of due dates appearing below. Details on the homework assignments, cribs, and points accumulated to date will be posted on the course website.

Assignment Schedule	Due Date & Time	Points
Homework 1	Jan 14 @ 4 pm EDT	30
Homework 2	Jan 28 @ 4 pm EDT	30
Assessment 1, Chap 14-17	Feb 3 @ 8:00-9:00 pm EDT in **PHYS 112/203**	150

Homework 3	Feb 11 @ 4 pm EDT	30
Homework 4	Feb 25 @ 4 pm EDT	30
Assessment 2, Chap 18-20	Mar 3 @ 8:00-9:00 pm EDT in LILY 1105	150
Homework 5	Mar 11 @ 4 pm EDT	30
Homework 6	Mar 25 @ 4 pm EDT	30
Assessment 3, Chap 21-22, Handout 1-3	Apr 7 @ 8:00-9:00 pm EDT in LILY 1105	150
Individual Synthesis Project	Apr 22 @ 4pm EDT	150
Comprehensive Final	TBD	300

- Homework Assignments (30 points each; the lowest score will be dropped from your total points accumulated). Six homework assignments will be given along with a deadline for uploading your responses to Brightspace. *No late homework assignments will be accepted – a zero will be recorded for that homework score if the deadline is missed.* The highest 5 scores on these assignments will be counted toward your point total for the course.
- Assessments (150 points each). There will be three in-class assessments, comprised of open response questions to evaluate your understanding of the course materials to date. *No make-up assessments will be possible – if one is missed, that will serve as the exam score that will be dropped from your total accumulated points.*
- Synthesis Project (150 points). A personalized target-based synthesis problem will be assigned in the middle of March that will require the use of reactions learned during the semester (and only those reactions) to solve the problem. You will record a 30-second video to describe your synthesis.
- Final Assessment (300 points). A comprehensive final assessment will be administered during Finals Week.
- The lowest hourly assessment score, or half the final assessment score, will be dropped from your total points accumulated before assigning your letter grade (≤ -150 points). For example, if you earned 150 points on each hourly assessment and 280 points on the final assessment, then 140 points – half the final assessment score – would be deducted from your point total. Similarly, if you earned 128 points on the first hourly assessment and perfect scores on all the other assessments, then 128 points would be deducted from your point total before letter grade assignment.

Total Points Possible

1. Homework (6 x 30 pts; 5 highest scores out of 6 assignments) 150 pts
2. Assessments (3 x 150 pts) 450 pts
3. Synthesis Project (150 pts) 150 pts
4. Comprehensive Final Assessment 300 pts
5. Drop Lowest Assessment Score (Assessments 1, 2, 3 or ½ Final) -150 pts

Total 1 – 4: 900 pts

Re-Grading Procedure

The following method must be followed for all regrades.

Step 1. Put in writing the question(s) you would like to have reconsidered and your rationale for why they should be regraded.

Step 2. Staple this written request to your graded assessment and give it to Ms. Jenkins.

Step 3. ***Steps 1 & 2 must be performed within 2 weeks of the assessment date to be eligible for a regrade.*** For Assessment 1, that deadline is Feb 10. For Assessment 2, the deadline is Mar 10. For Assessment 3, the deadline is Apr 14.

Step 4. The entire assessment will be re-graded and the points adjusted up or down as appropriate. The new score will be entered into the grade book.

Cheating

As a Boilermaker, your colleagues and I will expect you to live true to the Purdue Honors Pledge: *"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together – we are Purdue."* Students are required to do their own work on homework assignments and assessments. You may find it helpful to study together while preparing solutions to homework problems, however, you must not copy other's answers or use online homework solution resources – the answers must be written on your own without looking at someone else's paper, otherwise it will be treated as cheating. Anyone found cheating, for the first instance, will be given a zero and will NOT be allowed to drop the score. A second violation will result in immediate failure of the course.

Grading Scale

In keeping with University Senate regulations, all students will receive graded feedback in this course between the beginning of Week 5 and the end of Week 7. These grades ***will not*** become part of the permanent record; they are intended to provide guidance on your standing in the course at that time.

Guaranteed Grade Cut-offs (out of a total of 900 pts)

A	≥ 820
B	720 – 819
C	600 – 719
D	490 – 599
F	0 – 489

Academic Guidance in the Event when a student is Quarantined/Isolated

If you become quarantined or isolated at any point in time during the semester, in addition to support from the Protect Purdue Health Center, you will also have access to an Academic Case Manager who can provide you academic support during this time. Your Academic Case Manager can be reached at acmq@purdue.edu and will provide you with general guidelines/resources around communicating with your instructors, be available for academic support, and offer suggestions for how to be successful when learning remotely. Importantly, if you find yourself too sick to progress in the course, notify your academic case manager and notify me via email. Arrangements will be made based on your particular situation. The Office of the Dean of Students (odos@purdue.edu) is also available to support you should this situation occur. ***If you have any concerns about your health, contact the Protect Purdue Health Center at 765-496-4636.***

Attendance Policy during COVID-19

"If you feel ill, do not come to class, get tested for COVID and seek medical treatment if needed. Please provide hard copy evidence of testing to justify the absence and to be excused from that class. You will be excused from that laboratory with any documented test and it will not negatively affect your grade. This policy augments the existing Protect Purdue standard process of ODOS issuing notification of any student who is instructed by PPHC to isolate or quarantine. As always, please follow the Protect Purdue guidance found here https://protect.purdue.edu/protect-purdue-health-center/what-to-do-if-you-are-sick/". Consult the Brightspace shell for a link under the University Policies menu to review Attendance and Grief Absence policies.

Students should stay home and contact the Protect Purdue Health Center (496-INFO) if they feel ill, have any symptoms associated with COVID-19, or suspect they have been exposed to the virus. In the current context of COVID-19, the student still needs to inform the instructor of any conflict that can be anticipated and will affect the submission of a homework assignment or the ability to take an assessment. Only Professor Mei can excuse a student from a course requirement or responsibility. When conflicts can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency conflict, when advance notification is not possible, the student should contact Prof. Mei as soon as possible by email or by phone. When the student is unable to make direct contact with Prof. Mei and is unable to leave word with a course TA because of circumstances beyond the student's control, and in cases of bereavement, quarantine, or isolation, the student or the student's representative should contact the Office of the Dean of Students via email or phone at 765-494-7156.

Classroom Guidance Regarding Protect Purdue

The Protect Purdue Plan, which includes the Protect Purdue Pledge, is campus policy and as such all members of the Purdue community must comply with the required health and safety guidelines. Required behaviors in this class include: staying home and contacting the Protect Purdue Health Center (496-INFO) if you feel ill or know you have been exposed to the virus, wearing a mask in classrooms and campus building, at all times (e.g., no eating/drinking in the

classroom), disinfecting desk/workspace prior to and after use, maintaining proper social distancing with peers and instructors (including when entering/exiting classrooms), refraining from moving furniture, avoiding shared use of personal items, maintaining robust hygiene (e.g., handwashing, disposal of tissues) prior to, during and after class, and following all safety directions from the instructor.

Students who are not engaging in these behaviors (e.g., wearing a mask) will be offered the opportunity to comply. If non-compliance continues, possible results include instructors asking the student to leave class and instructors dismissing the whole class. Students who do not comply with the required health behaviors are violating the University Code of Conduct and will be reported to the Dean of Students Office with sanctions ranging from educational requirements to dismissal from the university.

Any student who has substantial reason to believe that another person in a campus room (e.g., classroom) is threatening the safety of others by not complying (e.g., not wearing a mask) may leave the room without consequence. The student is encouraged to report the behavior to, and discuss next steps with, their instructor. Students also have the option of reporting the behavior to the Office of the Student Rights and Responsibilities. See also Purdue University Bill of Student Rights.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds and is key quality for maintaining the value and respect that is engendered by recipients of a Purdue degree. It is also the basis for the ***Purdue Honor Pledge:*** *"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."*

In the event that potential breaches of this value are observed, individuals are encouraged to alert university officials by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Note regarding the use of commercial websites:

1. CHM 26200 course policy is that web pages like Course Hero, Quizlet or other online services are not appropriate to use for completing course work items such as homework assignments or assessments.
2. Since Prof. Mei's notes are copyright protected, they cannot be sold, bartered, or posted without his express written permission.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and

mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. The institution views, evaluates, and treats all persons in any University related activity or circumstance in which they may be involved, solely as individuals on the basis of their own personal abilities, qualifications, and other relevant characteristics.

Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. The University will conduct its programs, services and activities consistent with applicable federal, state and local laws, regulations and orders and in conformance with the procedures and limitations as set forth in Purdue's Equal Opportunity, Equal Access and Affirmative Action policy which provides specific contractual rights and remedies. Additionally, the University promotes the full realization of equal employment opportunity for women, minorities, persons with disabilities and veterans through its affirmative action program. Any question of interpretation regarding this Nondiscrimination Policy Statement shall be referred to the **Vice President for Ethics and Compliance** for final determination.

Accessibility

Purdue University and the instructors in this class strive to make the CHM 26100 learning experience as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, please contact Prof. Mei so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. More details are available on our course Brightspace under Accessibility Information. The University also has assistance available to help you make learning materials accessible. Some examples include:

- *Information from Innovative Learning on Universal Design for Learning*
- *Guidance from Innovative Learning on creating accessible documents*
- *Workshops on accessible materials suggested by the DRC*
- *Contact innovativelearningteam@purdue.edu with questions.*

Mental Health Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 - 5.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell.

Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. Questions? → Contact Purdue Wellness at evans240@purdue.edu.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours. A link to CAPS (listed as Purdue Counseling and Psychological Services) is also on the Brightspace template, under the Student Services and Resources section.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting Prof. Mei or the TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

CHM 26200: Tentative Lecture Schedule, Spring 2022

Date	Lect #	Chap	Topic
Jan 10	1	15	Molecular Orbitals of a conjugated system and allylic system
Jan 12	2	15	Diels-Alder Reaction
Jan 14	3	15	UV-Vis spectroscopy and colored compounds
Jan 17	MARTIN LUTHER KING JR. DAY (No Classes)		
Jan 19	4	16	Benzene and aromaticity
Jan 21	5	16	Huckel's Rule
Jan 24	6	16/17	Aromatic compounds and Electrophilic aromatic substitution
Jan 26	7	17	Effect of directing groups on electrophilic aromatic substitution
Jan 28	8	17	Nucleophilic aromatic substitution
Jan 31	9	14	Either, thioether and their reactions
Feb 2	10	14	Epoxide chemistry
Feb 3	Assessment 1, Chap 14-17, 8:00-9:00 pm EDT in PHYS 112/203		
Feb 4	11	14-17	Assessment-1 Overview
Feb 7	12	18	Carbonyl compounds, physical properties, and their industrial use
Feb 9	13	18	Synthesis of aldehyde and ketones
Feb 11	14	18	Chemical transformations of aldehyde and ketones
Feb 14	15	19	Introduction of Amines
Feb 16	16	19	Chemical transformations of amines
Feb 18	17	19	Synthesis of amines
Feb 21	18	20	Introduction to carboxylic acid
Feb 23	19	20	Synthesis of carboxylic acids
Feb 25	20	20	Chemical transformations of carboxylic acids
Feb 28	21	21	Introduction to carboxylic acid derivatives and the synthesis
Mar 2	22	21	Chemical transformations of carboxylic acid derivatives and
Mar 3	Assessment 2, Chap 18-20, 8:00-9:00 pm EDT in LILY 1105		
Mar 4	23		Assessment-2 Overview
Mar 7	24	22	Enols and Enolates
Mar 9	25	22	Condensations of carbonyl compounds
Mar 11	26	22	Alpha substitutions of carbonyl compounds
Mar 14-19	Spring Break		
Mar 21	27	Handout-1	Catalytic carbon-carbon bond formation - Introduction
Mar 23	28	Handout-1	Catalytic carbon-carbon bond formation – Allylic Alkylation
Mar 25	29	Handout-1	Catalytic carbon-carbon bond formation – Heck Reaction
Mar 28	30	Handout-2	Catalytic carbon-carbon bond formation – Negishi Coupling
Mar 30	31	Handout-2	Catalytic carbon-carbon bond formation – Suzuki Coupling
Apr 1	32	Handout-2	Catalytic carbon-carbon bond formation – Sonogashira

Apr 4	33	Handout-3	Catalytic carbon-carbon bond formation – Alkene Metathesis
Apr 6	34	Handout-3	Catalytic carbon-heteroatom bond formation – Buchwald-Hartwig
Apr 7	Assessment 3, Chap 21-22 and handout 1-3, 8:00-9:00 pm EDT in LILY 1105		
Apr 8	35		Assessment-3 Overview
Apr 11	36	26	Introduction to polymers
Apr 13	37	Handout-4	Polymerization mechanism
Apr 15	38	Handout-4	Polymer Characterizations
Apr 18	39	Handout-4	Living polymerization -ATRP
Apr 20	40	Handout-4	Living polymerization -RAFT
Apr 22	41	Handout-4	Living polymerization - ROMP
Apr 22	Individual Synthesis Project		
Apr 25	42	Handout-5	Supramolecular polymers
Apr 27	43	Handout-5	Semiconducting and conducting polymers
Apr 29	44	Handout-5	Plastic recycling and upcycling
May 2-7	Final Exam		

CHM 26200, 2022 Spring, Study Problems from Wade & Simek "Organic Chemistry", 9th Edition

Chapter	Suggested Problem Numbers
15	Page 759: 25, 27, 29, 30, 31, 33, 38, 39, 40, 41
16	Page 801: 31, 32, 33, 37, 42, 44, 45, 51
17	Page 870: 51, 54, 55, 57, 64, 66, 73, 78
14	Page 710: 32, 33, 39, 42, 46, 48, 52, 53
18	Page 931: 39, 48, 50, 57, 63, 66, 68, 69,
19	Page 992: 33, 34, 38, 40, 42, 46, 51, 60, 63
20	Page 1037: 31, 32, 33, 35, 39, 45, 48
21	Page 1103: 46, 48, 49, 50, 70
22	Page 1167: 61, 62, 63, 71, 75, 76, 77, 81, 85
24*	From Brown, Iverson, Anslyn and Foote, Organic Chemistry, 7th Edition
	Page 1044: 9, 15, 18, 20, 23, 25, 32 34
26	Page 1306: 21, 22, 23, 24, 25, 26, 27, 29, 33, 34